



Vivante VGLite Driver Quick Start Guide

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1 Quick Start VGLite Build and Test Guide

This document provides a quick overview and basic guidance for building and testing Vivante's VGLite driver software. For additional details, refer to the README files and example build scripts as well as the Vivante Driver Development Guide and other documentation provided.

2 VGLite Packages

The following table lists the packages needed for building the VGLite software.

Table 1. VGLite Packages

Package	Description	Notes
Vivante_VGLite_Src_drv_*.tgz	Contains the VGLite driver	Required by FPGA
Vivante_VGLite_Src_drv_RTOS-addon_*.tgz	Contains the VGLite RTOS driver	Required by RTOS Platform
Vivante_VGLite_Src_tst_*.tgz	Contains VGLite test cases, build scripts and README	Required by FPGA
Vivante_VGLite_Src_Toolkit_*.tgz	Contains VGLite SvgRender Toolkit code, build scripts and README	Required by FPGA

2.1 Script Usage to Decompress Driver Packages

The `decompress_tgz_files.sh` script is used to extract all compressed files from the driver package to a directory.

Usage:

```
./decompress_tgz_files.sh -p <Source package directory> -o <Destination directory>
```

Refer to the following table for details about the options available for this command.

Table 2. decompress_tgz_file.sh Options

Option	Required or Not	Description
-p < Source package directory >	Required	Directory for storing Vivante_*.tgz
-o< Destination directory >	Required	Output directory
-h	Optional	Displays help information.

Note: The path of the destination directory in this command is referred to as the extraction path.

2.2 Environment Requirements

- Ubuntu 16.04 or later
- GCC 5.4.0 or later

3 Build VGLite Driver/util/Case/Toolkit on FPGA

The following steps build the VGLite Driver/util/Case/Toolkit on FPGA. The FPGA environment is Ubuntu 16.04_x86_64 and kernel 4.8.0 is used in this example. The build options may be changed corresponding to your environment.

- Step 1. Install all packages to your \$wksp. Refer to “Script Usage to Decompress Driver Packages” to extract all packages.
- Step 2. Go to the VGLite Driver directory: \$wksp/VGLite_driver.
- Step 3. Execute following command, it will list the build usage.

```
./build_linux.sh -h
```

Please use the print out Usage to build the VGLite driver.
- Step 4. Go to the VGLite Case directory: \$wksp/util
- Step 5. Build the VGLite util with the following commands:

```
./build_sample.sh clean 265 PCIE-GEN6
./build_sample.sh install 265 PCIE-GEN6
```
- Step 6. Go to the VGLite Case directory: \$wksp/Test_cases
- Step 7. Build the VGLite Case with the following commands:

```
./build_sample.sh clean 265 PCIE-GEN6
./build_sample.sh install 265 PCIE-GEN6
```
- Step 8. Go to the VGLite SvgRender Toolkit directory: \$wksp/SvgVGLiteRenderer.
- Step 9. Build the VGLite SvgRender Toolkit with the following commands:

```
cd $wksp/SvgVGLiteRenderer/freetype
./build_freetype_x86.sh
cd $wksp/SvgVGLiteRenderer
./build_sample.sh clean 265 PCIE-GEN6
./build_sample.sh install 265 PCIE-GEN6
```
- Step 10. The VGLite Driver/util/Case/ToolKit builds in the following subdirectories at \$wksp/build.s2c.
VGLite Driver: \$wksp/build.s2c/sdk/drivers
VGLite util: \$wksp/build.s2c/sdk/drivers
VGLite Case: \$wksp/build.s2c/sdk/samples
VGLite Toolkit: \$wksp/build.s2c/sdk/SvgVGLiteRenderer
- Step 11. Give write permission for the VGLite Driver/util/Case/Toolkit path.

4 Test VGLite Case/Toolkit on FPGA

The following steps outline the test procedure for the VGLite Case/Toolkit on FPGA. The FPGA environment is Ubuntu 16.04_x86_64 and kernel 4.8.0 is used in this example. The build options may be changed corresponding to your environment.

- Step 1. Go to the VGLite Driver build out path: `$wksp/build.s2c/sdk/drivers`.
- Step 2. Load the VGLite Kernel Object module. Example: `insmod vg_lite.ko ...`
- Step 3. Set environment variables to find the VGLite Cases required libraries.
Example: `export LD_LIBRARY_PATH=`pwd``.
- Step 4. Go to the VGLite Case build out path: `$wksp/build.s2c/sdk/samples`.
- Step 5. Execute the build out VGLite Case Binaries to generate images.
Example `./tiger`. This will generate one image named `tiger.png`, then check the generated rendered image.
Note: Before running `ft255/unit_test/unit_test1`, you will need to create a directory for the image results: `mkdir result_bmp`. Then these three cases will generate images that will save to the `result_bmp` directory.
Go to the VGLite Toolkit build out path: `$wksp/build.s2c/sdk/SvgVGLiteRenderer`.
- Step 6. Execute the build out VGLite Toolkit Binary to verify the online `SvgVGLiteRenderer` function.
Example: `./main circle.svg`. This will generate the corresponding image, then check the generated rendered image.
Note: The `svg` files at `$wksp/build.s2c/sdk/SvgVGLiteRenderer/example/vivante`.

Document Revision History

This section describes top level differences between document revisions.

Document Revision	Date	Compatible SW Release	Description
1.00	2026-02-03	VGLite from February 2026	Convert from Programming Memo to Driver Quickstart document. Section 2 VGLite Packages: updated Table 1. VGLite Packages. Added Section 2.1 Script Usage to Decompress Driver Packages. Section 3 Build VGLite Driver/util/Case/Toolkit on FPGA: updated steps. Section 4 Test VGLite Case/Toolkit on FPGA: update directories in test procedure steps.
0.80	2025-11-12	VGLite from November 2021	Initial release.

